DS	**Session**	Tree 1	**Question Information**	Level 1 Challenge 67

Question description

The sam is enjoying a wonderful vacation in Byteland. What makes the sam impressed the most is the road system of the country. Byteland has N cities numbered 1 through N. City 1 is the capital of Byteland. The country also has N-1 bidirectional roads connecting the cities. The i-th road connects two different cities ui and vi. In this road system, people can travel between every pair of different cities by going through exactly one path of roads.

The roads are arranged in such a way that people can distinguish two cities only when both cities have different number of roads connected to it. Such two cities will be considered similar. For example, city A is similar to the capital if the number of roads connected to city A is equal to the number of roads connected to the capital.

On each day during the vacation, the sam wants to have a trip as follows. He chooses two cities A and B such that the sam will visit city B if he goes from A to the capital using the shortest path. Then, the sam will visit the cities on the shortest path from A to B through this path. Please note that A may be equal to B; that means the sam will enjoy the day in a single city.

The sam does not want to have similar trips. Two trips are considered similar if and only if
they both have the same number of visited cities and for each i, the i-th city visited in one trip is similar to the i-th city visited in the other trip.

The sam wants to have as many different, namely not similar, trips as possible. Help him count the maximum number of possible trips such that no two of them are similar.

Input Format
The first line of the input contains a single integer N. The i-th line of next N-1 lines contains two space-separated integers ui and vi, denoting the i-th road.

Output Format
Output a single line containing the maximum number of different trips.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
2 1
3 1
```

EXPECTED OUTPUT

```
3
```

Test Case 2

INPUT (STDIN)

```
4
2 1
3 1
2 4
```

EXPECTED OUTPUT

```
5
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
struct state
```

Test Case 2

KEYWORD

```
const int MAXL=200005;
```

Test Case 3

KEYWORD

```
state st[MAXL]
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
1
```

Test Case 2

TOKEN COUNT

```
700
```

Test Case 3

NLOC

```
67
```